

SEEMAB, CLASS 9 (FEDERAL BOARD)
Biology exam-17 (Version B): Detailed Marking Report

Chapters 1 to 7 (complete) | 38 marks, no choices | attempted and marked 17 September 2026 | independently deep-checked

36/38 TOTAL	94.7% PERCENTAGE	A+ GRADE	10/10 SECTION A	11/12 SECTION B	15/16 SECTION C
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The short version. Seemab scored 36 out of 38 (94.7%, A+) on the book-style Version B that she asked for herself. The MCQs were perfect, the light dependent reactions answer (sketch and all five book steps) was full marks, and both Chapter 3 and Chapter 7 short questions on plastids and respiration were complete. The 2 marks lost are two equation slips and one difference that answered about meiosis as a whole instead of meiosis II.

SECTION A: Multiple Choice (10 / 10)

Q	Topic	Hers	Key	Note
1	Starting point of an investigation (Ch1 book MCQ 7)	C	C	Reason given: hypothesis comes after questioning, data is a gathered result, theory is a well defined statement.
2	Information from an experiment (Ch1 book MCQ 8)	D	D	"theory, observation and hypothesis aren't suitable".
3	Same class, same phylum (Ch2 book MCQ 17)	A	A	"All other options aren't suitable".
4	Thylakoids fuse to form intergrana (Ch3)	B	B	Elimination reason.
5	Vitamin D is a lipid (Ch6 book MCQ 10)	B	B	Elimination reason.
6	Brain uses 25% of ATP (Ch7)	A	A	Letter only.
7	Light independent reactions (Ch7)	A	A	Letter only.
8	FADH ₂ gives two ATP (Ch7)	C	C	First letter fully blotted out, clean C written beside it: a proper correction.
9	Yeast: wine making and baking (Ch7)	D	D	Letter only.
10	Oxygen + H ⁺ gives water (Ch7)	B	B	Letter only.

Every answer numbered 1 to 10 as on the paper, and every number matches its letter (shift guard clean). She gave written reasons for the first five.

SECTION B: Short Questions (11 / 12)

Q.2(i) Equations: 3 / 4

- (a) $6\text{CO}_2 + 12\text{H}_2\text{O} \xrightarrow{\text{light, chlorophyll}} 6\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{H}_2\text{O} + 6\text{O}_2$
 (b) Pyruvic acid ----> Ethyl + CO₂
 (c) $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \text{ ----> } 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{energy (ATP)}$
 (d) Lactic acid.

- (a) 0.5: everything matches the book (12H₂O, 6H₂O, 6O₂, light and chlorophyll) except the "6" written in front of glucose. At 300 and 600 DPI it is a complete, separate 6, not a retraced C, so the equation makes six glucose and the carbon does not balance.
 (b) 0.5: "Ethyl" on its own is not a compound; the book product is ethyl alcohol. (c) 1 after a scribbled-out term. (d) 1.

Q.2(ii) Unripe and ripe oranges: 4 / 4

"Green colour is given by chloroplast." "Orange colour is given by chromoplasts." "Chromoplasts are also found in flowers. They give flowers their bright colour to attract pollinators." "help to attract animals for seed dispersal." All four parts as the book states.

Q.2(iii) Aerobic vs anaerobic respiration: 4 / 4

Four rows: "requires oxygen / does not require oxygen"; "Produces more ATP / Produces less ATPs"; "Glucose is completely broken down / partially broken down"; "End products are CO₂ and H₂O / may be lactic acid or ethanol + CO₂ (yeast)". She added both equations as well.

SECTION C: Long Questions (15 / 16)

Q.3(i) Light dependent reactions, describe and sketch: 8 / 8

Sketch 3/3: photosystem II and I, photons, H₂O splitting to 2H⁺ and ½O₂, 2e⁻, primary electron acceptor, electron transport chain, "energy for synthesis of ATP", NADP⁺. Description 5/5: she opened with thylakoid membranes and the two photosystems, then wrote all five book steps almost word for word, from "Chlorophyll a of photosystem II absorbs light" to "reduce NADP⁺ to NADPH" (the fifth step is numbered 3, a numbering slip only).

Q.3(ii) Meiosis II is identical to mitosis: 7 / 8

- (a) 4/4: prophase II "Chromosomes condense, the spindle forms and the nuclear envelope breaks down"; metaphase II line up at the equator with spindle fibres attached; anaphase II "sister chromatids separate and are pulled to the opposite poles"; telophase II reach the poles, envelope reforms, chromosomes uncoil.
 (b) 2/2: sister chromatids separate during anaphase; chromosomes line up at the equator during metaphase and spindle fibres are involved.
 (c) 1/2: her table is headed "Mitosis / Meiosis". "occurs in somatic cells / occurs in germ line cells" earns 1. "cells divides only once / cell divide twice" earns 0: that is a difference between mitosis and meiosis as a whole, but the question compares mitosis with meiosis II, which is itself a single division. Differences that would score: meiosis II starts from haploid cells, there is no DNA replication before it, and its products are not genetically identical because of crossing over.

Summary of lost marks

Where marks went	Marks	Kind
2(i)(a) photosynthesis equation	0.5	Wrote 6C ₆ H ₁₂ O ₆ : six glucose instead of one, so carbon does not balance

2(i)(b) alcoholic fermentation	0.5	"Ethyl + CO ₂ ": the word alcohol missing (she wrote it correctly in 2(iii))
3(ii)(c) meiosis II vs mitosis	1	"divides only once / divide twice" compares mitosis with the whole of meiosis; meiosis II is itself one division

Marking method

Marked from the 7-page CamScanner scan at 110 DPI, with MCQ 8 and the three equations re-read at 300 DPI. An independent background re-mark re-derived all ten MCQ answers, ran a shift guard, and re-marked every part without seeing the first pass. The first pass gave 36.5; the re-mark read the glucose "6" at 600 DPI as a deliberate coefficient and took half a mark, which was accepted. Final 36/38.

END OF REPORT